

Db2 System Requirements, Installation & Interfaces -- DBMS Module 6

Definition: Setting up IBM Db2 requires verifying operating system and hardware prerequisites, performing installation via IBM DB2 Setup Wizard, and interacting with the database using CLP (Command Line Processor), IBM Data Studio, or Db2 Web Console.

1. Detailed Technical Explanation

1. System Requirements for IBM Db2 (v11.5)

- * Supported OS: Red Hat Enterprise Linux (RHEL), SUSE Linux, Ubuntu, Windows Server / Windows 10/11 (64-bit), IBM AIX.

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2. Step-by-Step Db2 Installation Process

```
1. Download Db2 Installation Package (e.g., v11.5 Community Edition)
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2. Run Setup Executable (setup.exe on Windows, ./db2setup on Linux)
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3. Select Installation Type (Typical / Compact / Custom)
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4. Set Up Db2 Instance (Default instance: DB2 / db2inst1)
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5. Configure Administrator Credentials (Username: db2admin / db2inst1)
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6. Complete Installation & Verify Service Status via Command Line
```

3. IBM Db2 Management Interfaces

1. Db2 Command Line Processor (CLP)

The primary command-line tool for executing SQL statements, system commands, and database management utilities.

```
# Start Db2 Command Line Processor
db2cmd

# Connect to database SAMPLE
db2 CONNECT TO SAMPLE USER db2admin USING password123

# Execute SQL Query via CLP
db2 "SELECT * FROM EMPLOYEE WHERE DEPTNO = 'D11'"

# Disconnect from database
db2 CONNECT RESET
```

2. IBM Data Studio

An Eclipse-based GUI tool for database administration, SQL query building, stored procedure debugging, and schema modeling.

3. Db2 Console / Web Console

A modern browser-based web dashboard providing real-time database monitoring, performance alerts, memory usage metrics, and query

execution execution plan visualizers.

4. Core Concepts & Memory Keywords

- * Db2 Instance: A logical execution environment that manages database files, memory buffers, and configuration parameters.

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5. Must-Write Points for Exams

- * A Db2 Instance must be created and started (db2start) before creating or connecting to databases.

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6. Quick Recall Flow

System Check -> Run db2setup -> Create Instance (db2inst1) -> Start Service (db2start) -> Interact via CLP / Data Studio